Project 1

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Contents

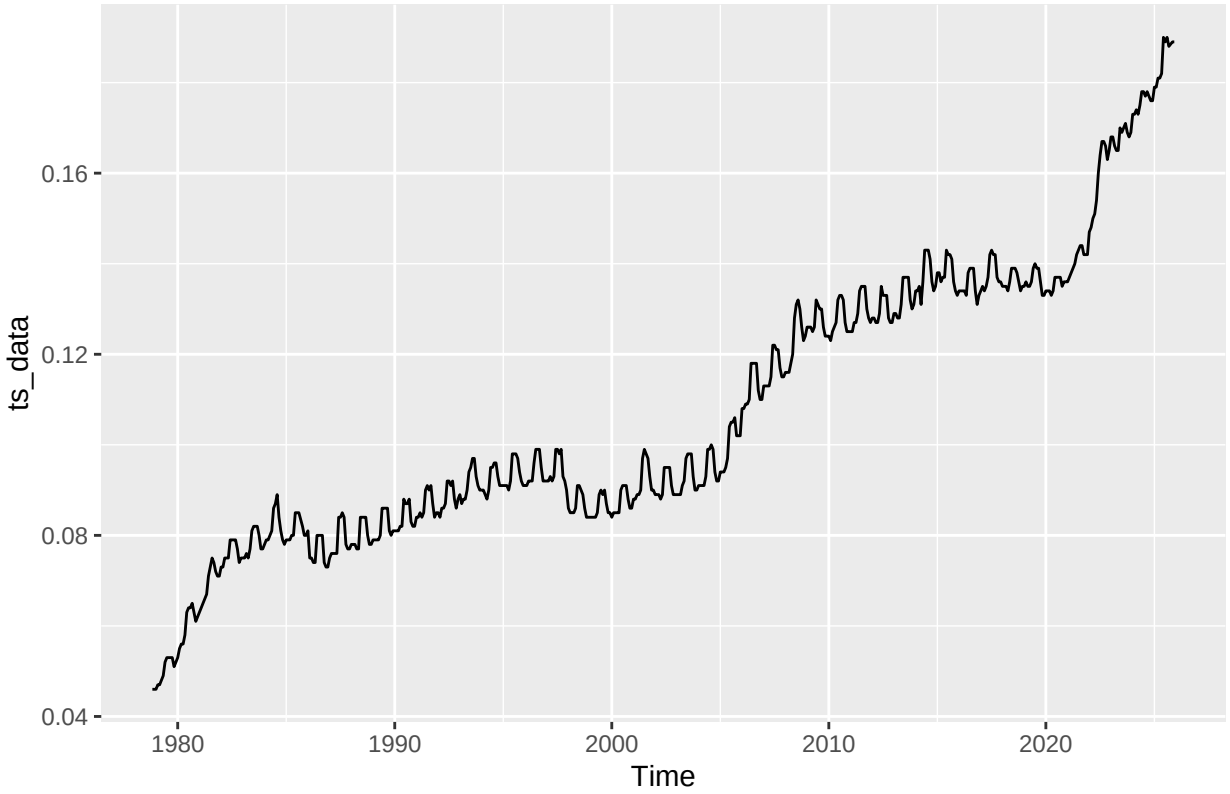
Introduction

This project analyzes the monthly time series APU000072610 from the Federal Reserve Economic Data (FRED), which measures U.S. residential electricity prices for urban consumers. The data starts in November 1978 and runs through December 2025. Electricity prices fluctuate around the year due to demand fluctuations and seasonal factors such as generation costs and regulation making it fit for data with relevant seasonality. Electricity prices are also expected to grow over time meaning that there should also be a trend component.

Results

1a)

```
data <- read_csv("../../data/APU000072610.csv")
ts_data <- ts(data[2], start=c(1978, 11), frequency=12)
ts_data <- na.approx(ts_data)
autoplot(ts_data)
```

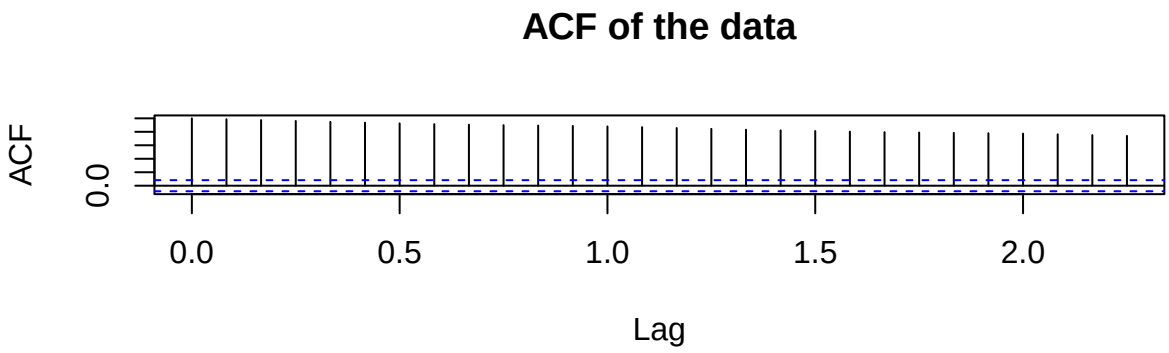


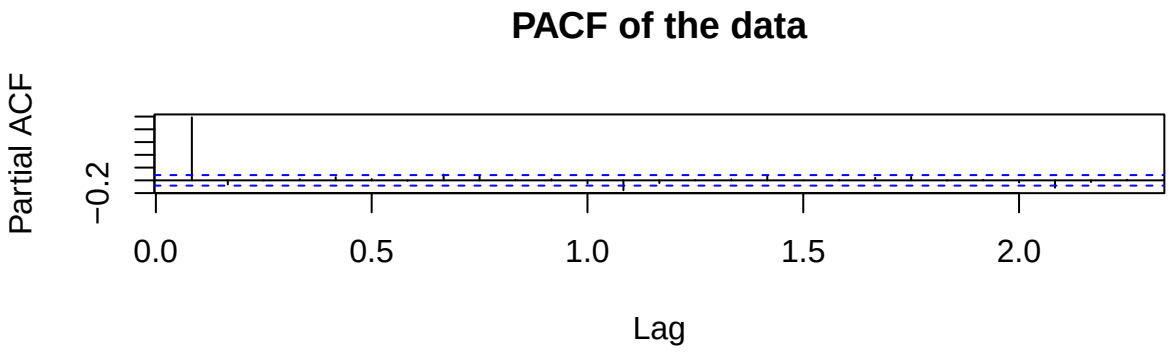
1b)

The plot in part a does not suggest that the data is covariance stationary. There is a clear upward trend which means that the time series is not mean reverting.

1c)

```
par(mfrow=c(2,1))
acf(ts_data, main="ACF of the data")
pacf(ts_data, main="PACF of the data")
```



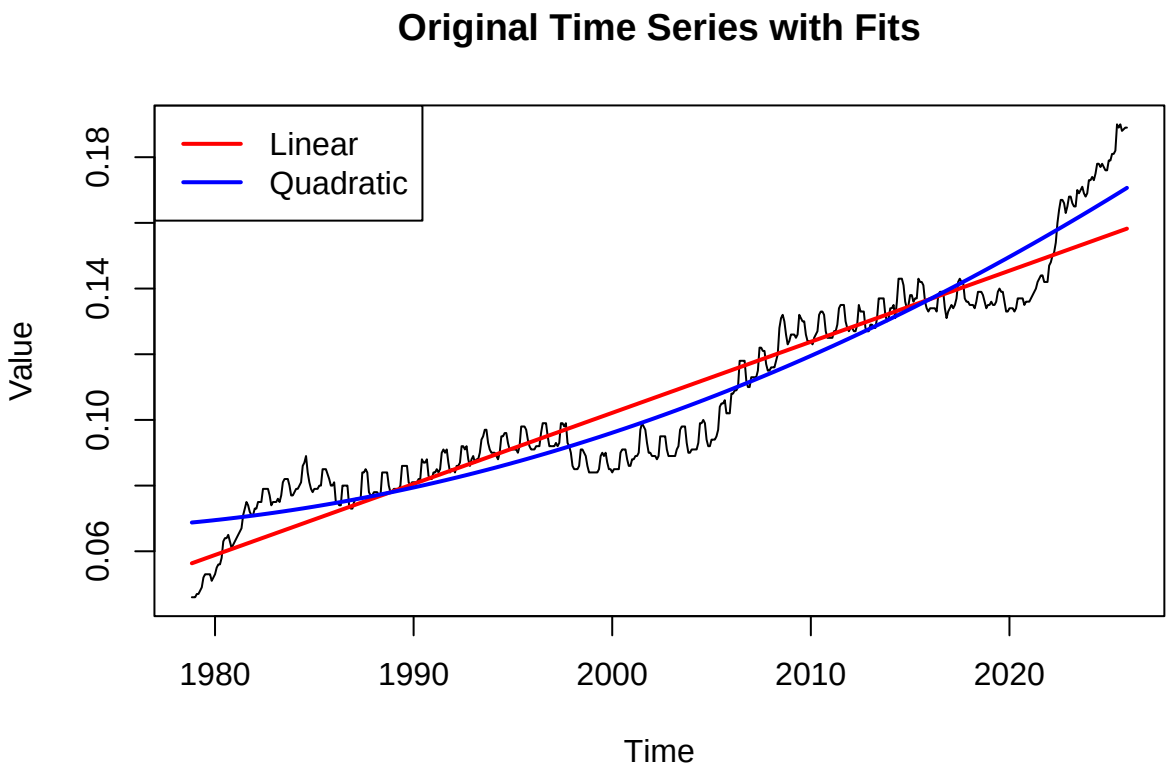


The ACF of the data is very significant with a declining lag, this suggests autocorrelation. The PACF has some significant spikes, but mostly below the line for statistical significance. This suggests that the data is likely driven by an autoregressive component.

1d)

```
t <- time(ts_data)
model1 <- lm(ts_data ~ t)
model2 <- lm(ts_data ~ t + I(t^2))

plot(ts_data, main="Original Time Series with Fits", ylab="Value", xlab="Time")
lines(ts(fitted(model1), start=start(ts_data), frequency=frequency(ts_data)), col="red", lwd=2)
lines(ts(fitted(model2), start=start(ts_data), frequency=frequency(ts_data)), col="blue", lwd=2)
legend("topleft", legend=c("Linear", "Quadratic"), col=c("red", "blue"), lwd=2)
```

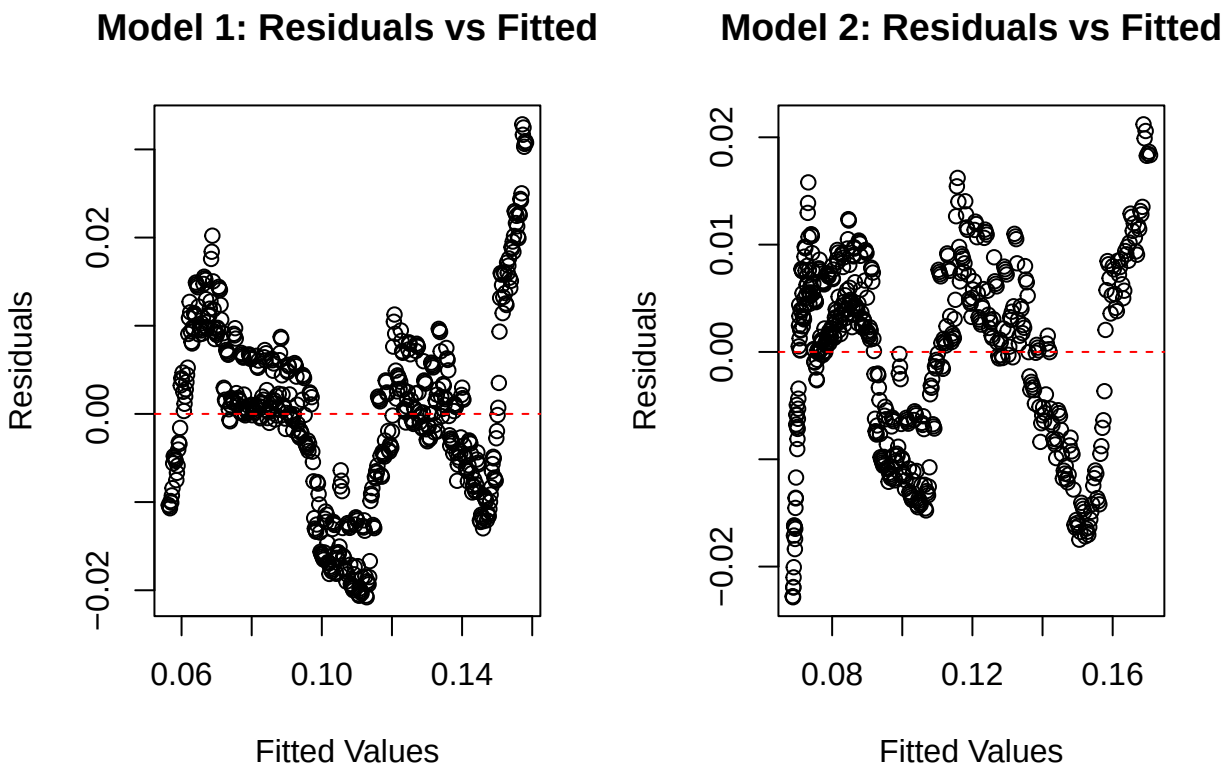


1e)

```
par(mfrow=c(1,2))

# Model 1: Residuals vs Fitted
plot(fitted(model1), residuals(model1),
     main="Model 1: Residuals vs Fitted",
     xlab="Fitted Values", ylab="Residuals")
abline(h=0, col="red", lty=2)

# Model 2: Residuals vs Fitted
plot(fitted(model2), residuals(model2),
     main="Model 2: Residuals vs Fitted",
     xlab="Fitted Values", ylab="Residuals")
abline(h=0, col="red", lty=2)
```

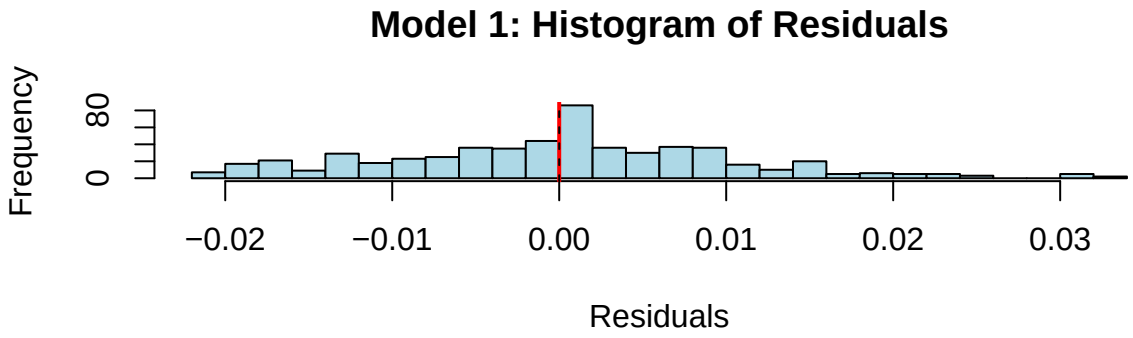


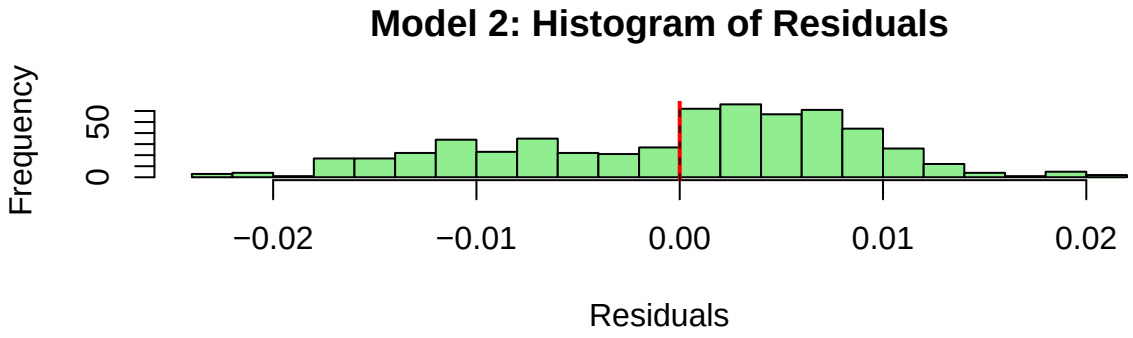
There is a very evident pattern in the residuals vs fitted models, this suggests that the residuals are not constant over time and that there is some sort of dynamic present in the data. This implies that the trend-only model fails to account for the entirety of the data-generating process, likely due to seasonality.

1f)

```
# Model 1: Histogram of residuals
par(mfrow=c(2,1))
hist(residuals(model1),
     main="Model 1: Histogram of Residuals",
     xlab="Residuals",
     ylab="Frequency",
     breaks=20,
     col="lightblue")
abline(v=mean(residuals(model1)), col="red", lwd=2)
abline(v=0, col="black", lty=2)

# Model 2: Histogram of residuals
hist(residuals(model2),
     main="Model 2: Histogram of Residuals",
     xlab="Residuals",
     ylab="Frequency",
     breaks=30,
     col="lightgreen")
abline(v=mean(residuals(model2)), col="red", lwd=2)
abline(v=0, col="black", lty=2)
```





Here, we can see that in model 1 the residuals follow more of a normal distribution, although they are more spread out than the residuals of model 2. For model 2, there is a left-skew in the residuals, this indicates that even though the quadratic trend follows the path graphically, there is still unmodeled variation in the data.

1g)

```
summary(model1)
```

```
##
## Call:
## lm(formula = ts_data ~ t)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.020792 -0.006684  0.000466  0.006358  0.032840
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.226e+00  6.365e-02  -66.40   <2e-16 ***
## t            2.164e-03  3.179e-05   68.08   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0103 on 564 degrees of freedom
## Multiple R-squared:  0.8915, Adjusted R-squared:  0.8913
## F-statistic:  4635 on 1 and 564 DF,  p-value: < 2.2e-16
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = ts_data ~ t + I(t^2))
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.022856 -0.006633  0.001734  0.006535  0.021208
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.308e+02  8.795e+00   14.87   <2e-16 ***
## t           -1.327e-01  8.785e-03  -15.10   <2e-16 ***
## I(t^2)       3.367e-05  2.194e-06   15.35   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.008654 on 563 degrees of freedom
## Multiple R-squared:  0.9235, Adjusted R-squared:  0.9232
## F-statistic:  3399 on 2 and 563 DF,  p-value: < 2.2e-16
```

```
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: ts_data ~ t
## Model 2: ts_data ~ t + I(t^2)
##   Res.Df      RSS Df Sum of Sq      F    Pr(>F)
## 1    564 0.059802
## 2    563 0.042163  1   0.01764 235.54 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the regression summaries for model 1 and model 2, the trend terms are significant from their t-tests and the overall models are both significant from their F-tests. From the adjusted R^2, the quadratic model is able to better explain variation in the data, however, the difference is not drastic. After running anova, model 2 is preferred over model 1 as it improves the fit at a statistically significant level. However, due to the residuals still showing structure rather than random noise, these statistics should be taken with caution as their precision may be jeopardized by correlation and heteroskedasticity in the errors.

1h)

```
AIC(model1, model2)
```

```
##        df       AIC
## model1  3 -3569.663
## model2  4 -3765.483
```

```
BIC(model1, model2)
```

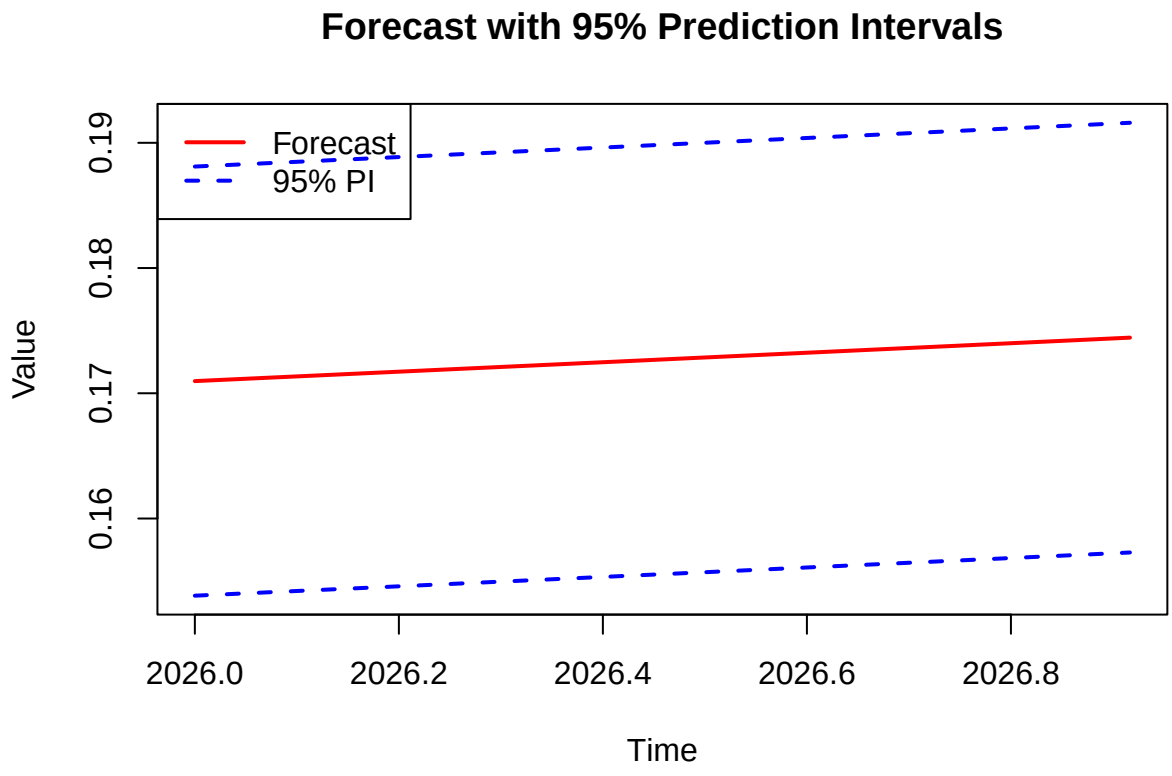
```
##        df       BIC
## model1  3 -3556.647
## model2  4 -3748.129
```

From the AIC and BIC, model2 is preferred over model1 because it has a lower value for both metrics. AIC and BIC both agree on what the selected model should be.

1i)

```
future_t <- seq(max(t) + 1/12, by = 1/12, length.out = 12)
tn <- data.frame(t = future_t)
tn$`I(t^2)` <- tn$t^2
pred.plim <- predict(model2, tn, level = 0.95, interval = "prediction")

plot(future_t, pred.plim[, "fit"], type = "l", col = "red", lwd = 2,
     ylim = range(pred.plim), main = "Forecast with 95% Prediction Intervals",
     ylab = "Value", xlab = "Time")
lines(future_t, pred.plim[, "lwr"], col = "blue", lty = 2, lwd = 2)
lines(future_t, pred.plim[, "upr"], col = "blue", lty = 2, lwd = 2)
legend("topleft", legend = c("Forecast", "95% PI"),
       col = c("red", "blue"), lty = c(1, 2), lwd = c(2, 2))
```

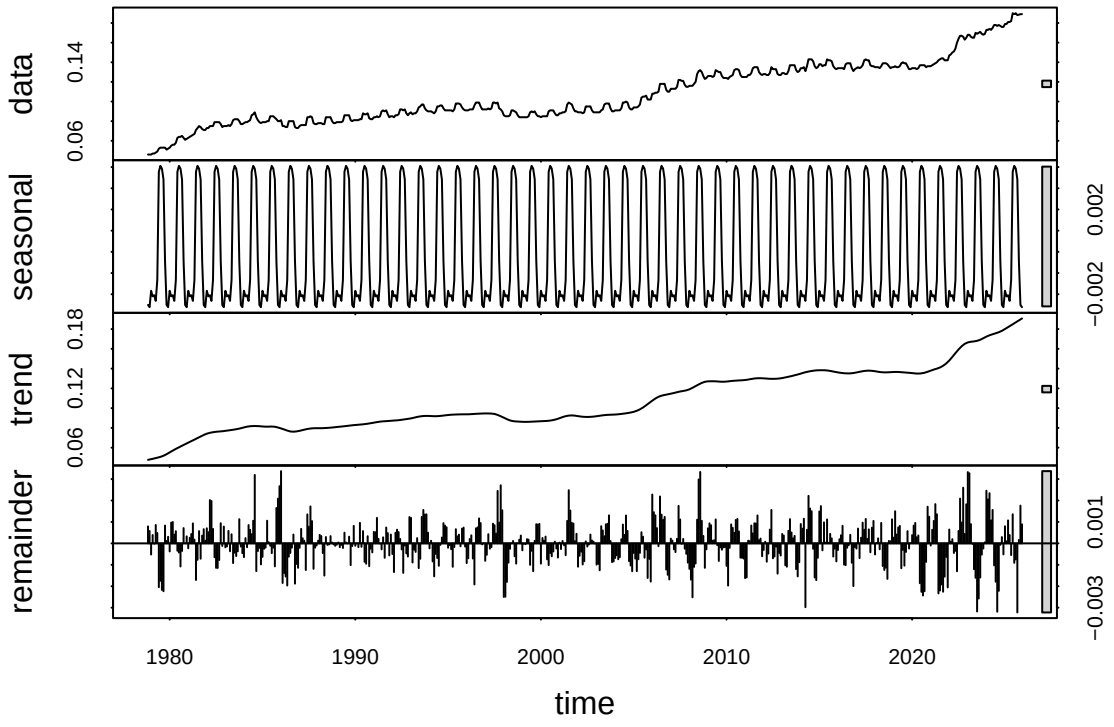


Part 2

2a)

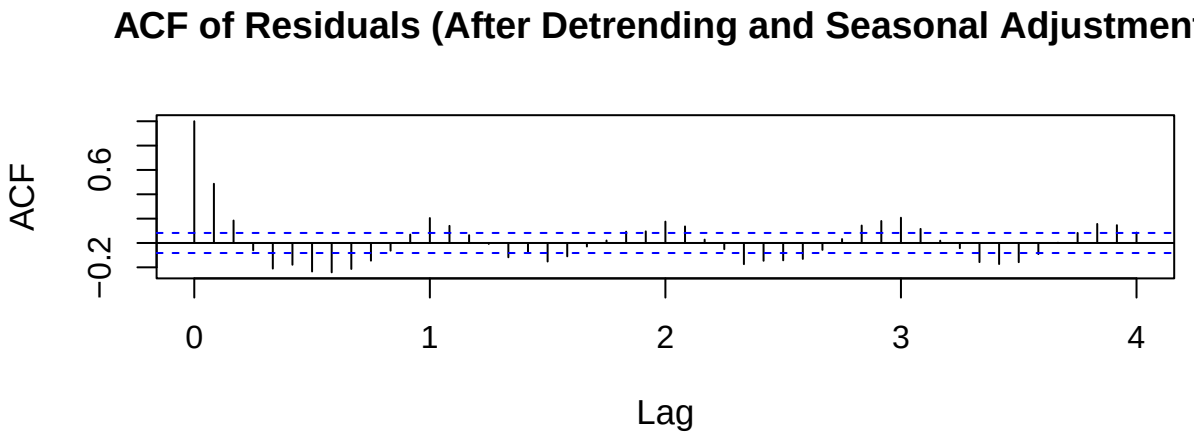
```r
# Convert to univariate (extract as numeric vector)
ts_data_uni <- ts(as.numeric(ts_data), start = c(1978, 11), frequency = 12)

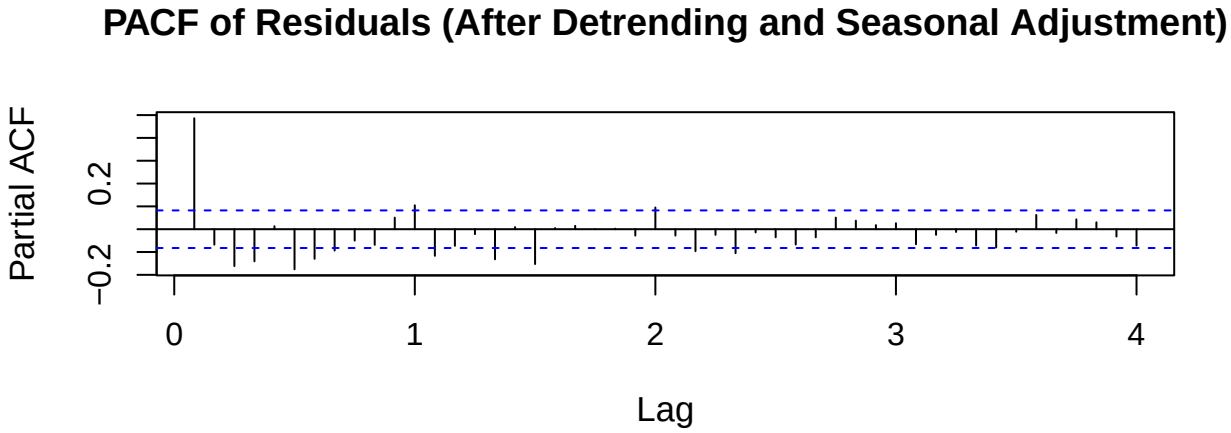
# Now STL will work
decomp <- stl(ts_data_uni, s.window = "periodic")
plot(decomp)
```




```r
# Extract residuals
residuals_decomp <- decomp$time.series[, "remainder"]

# Plot ACF and PACF
par(mfrow = c(2, 1), mar = c(4, 4, 3, 2), cex.main = 1.2, cex.lab = 1.1)
acf(residuals_decomp, main = "ACF of Residuals (After Detrending and Seasonal Adjustment)",
    lag.max = 48, cex.main = 1.2)
pacf(residuals_decomp, main = "PACF of Residuals (After Detrending and Seasonal Adjustment)",
     lag.max = 48, cex.main = 1.2)
```



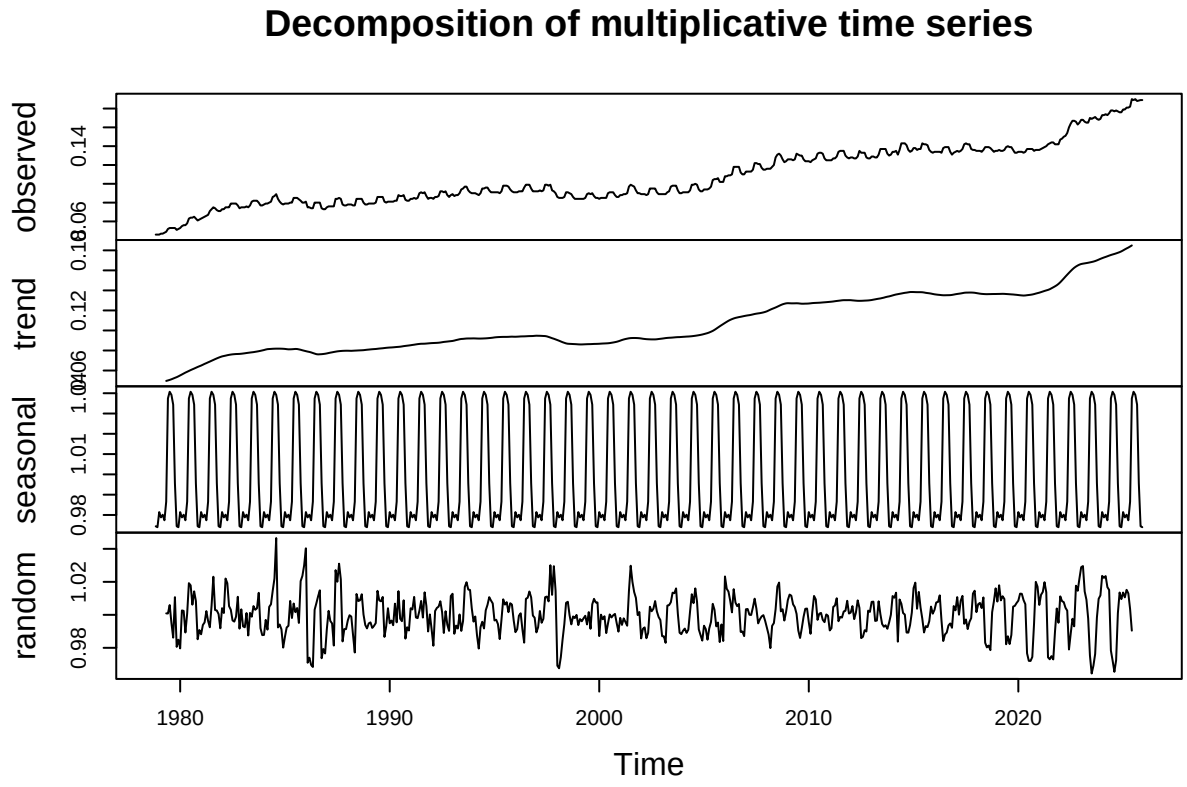


The additive decomposition separated the time series into a trend, seasonal, and error component. The trend had a smooth upward, slightly quadratic trajectory. The seasonal component had a consistent monthly pattern, and the remainder is centered near zero and has no visible pattern, but it is better to reference the ACF and PACF before making a conclusion.

The ACF and PACF show some autocorrelation at later lags and follow a pattern that looks like a cosine wave, this could indicate that there is another factor such as cycles that is introducing further dynamics in the random component and affecting the residuals.

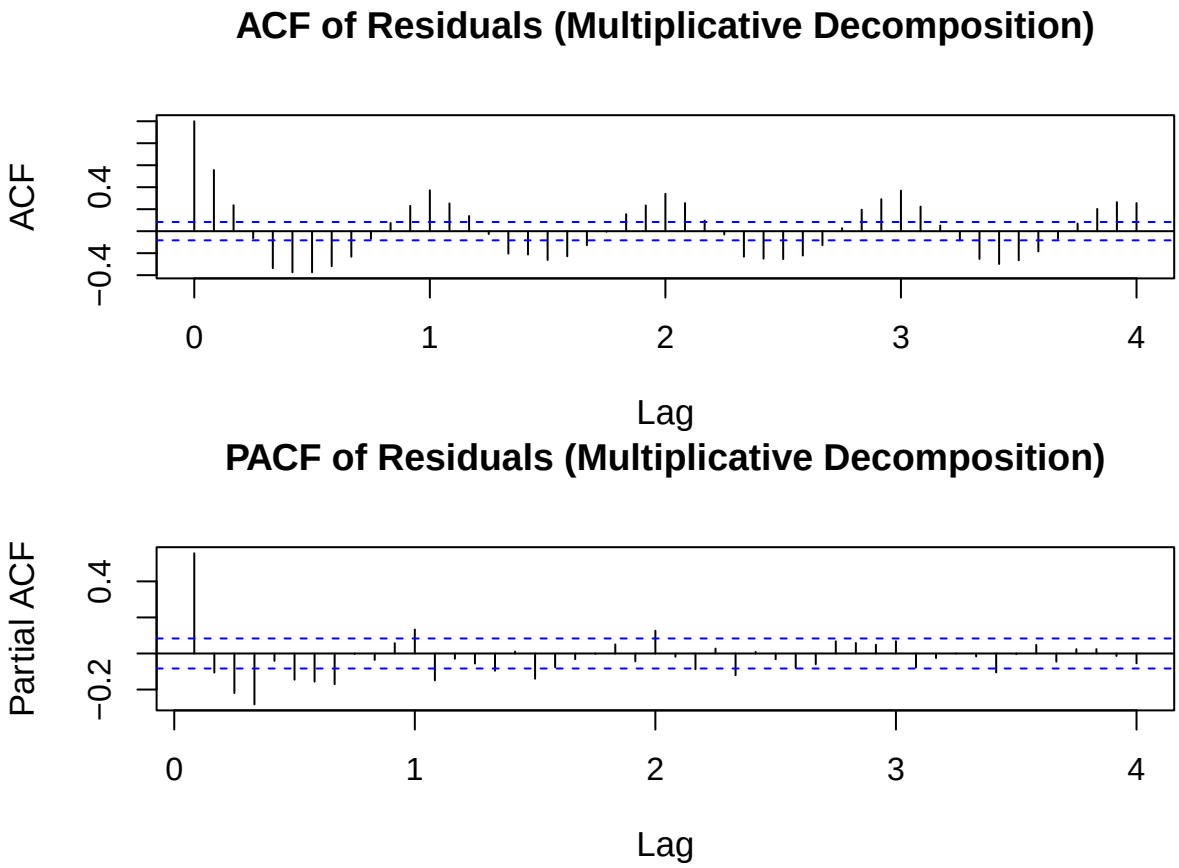
2b)

```
# Alternative: Use decompose() for multiplicative
decomp_mult <- decompose(ts_data_uni, type = "multiplicative")
plot(decomp_mult)
```




```r
# Extract residuals
residuals_decomp_mult <- decomp_mult$random  # "random" is the remainder in decompose()

# Plot ACF and PACF
par(mfrow = c(2, 1), mar = c(4, 4, 3, 2), cex.main = 1.2, cex.lab = 1.1)
acf(na.omit(residuals_decomp_mult), main = "ACF of Residuals (Multiplicative Decomposition)",
    lag.max = 48, cex.main = 1.2)
pacf(na.omit(residuals_decomp_mult), main = "PACF of Residuals (Multiplicative Decomposition)",
     lag.max = 48, cex.main = 1.2)
```



The multiplicative decomposition of the time series is somewhat similar to the additive. Here, the seasonality is seen as a proportion to the data, however, the original series did not suggest a need for multiplicative seasonality as the amplitude of the seasonality did not change as the y-value grew. The results are similar with a clear seasonal component, a smooth upward trend, and a noisy random component indicative of white noise.

Similar to the additive, the ACF and PACF have a pattern which would indicate that there are still dynamics in the system that are still present in the random component, it has a period of 1 year, so this could likely be additional seasonal dynamics.

2c)

Since the amplitude of the data is stable over time and does not grow with the y-value increasing, the additive decomposition is preferred in this case. This can also be seen in the ACF and PACF where the additive for these two exhibited lower autocorrelation compared to those of the multiplicative decomposition.

2d)

The models for the cycles would be somewhat similar, however, it would be tougher and more complex for the multiplicative model because the trend and seasonality was not as effectively removed compared to the additive model. Since there is more residual dependence in the multiplicative, there would be a bigger cycle structure for that model.

2e)

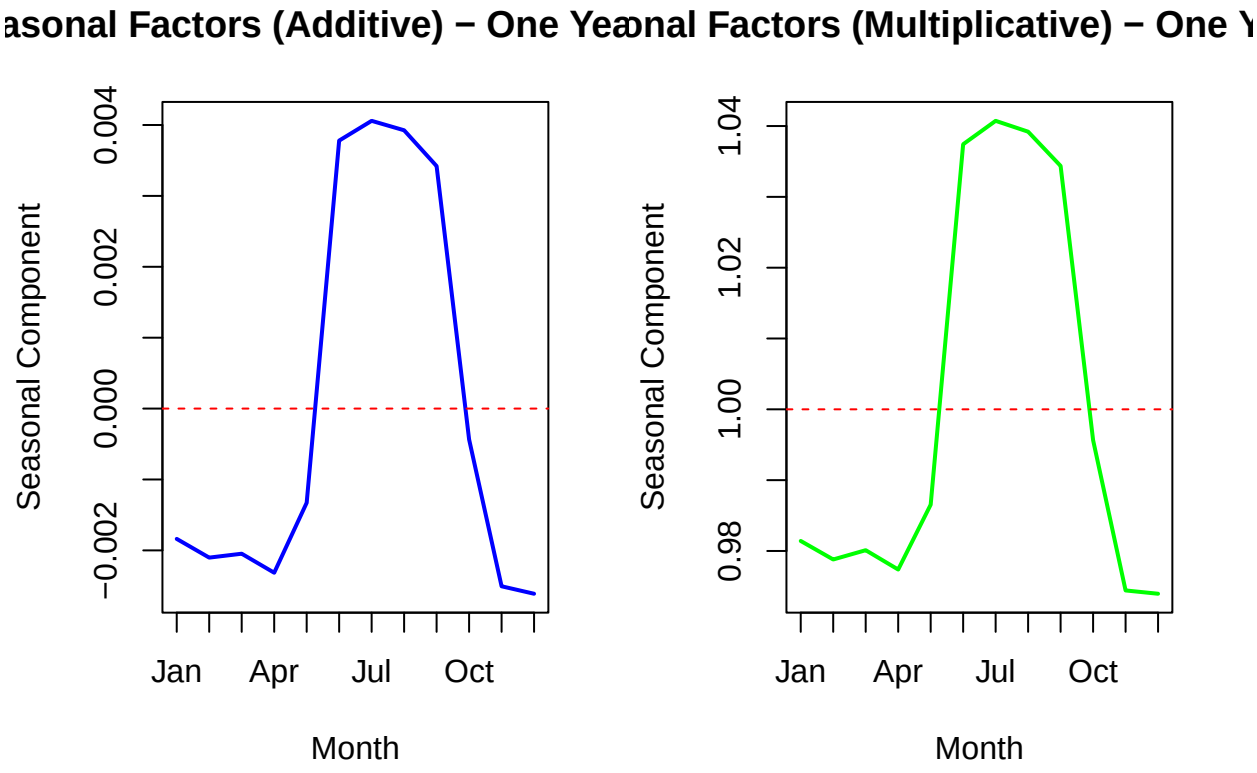
```
# Extract seasonal components from both decompositions
seasonal_add <- decomp$time.series[, "seasonal"]
seasonal_mult <- decomp_mult$seasonal
```

```r
# Extract one yearly cycle, January to December (the series starts in November 1978)
seasonal_add_cycle <- as.numeric(window(seasonal_add, start = c(1979, 1), end = c(1979, 12)))
seasonal_mult_cycle <- as.numeric(window(seasonal_mult, start = c(1979, 1), end = c(1979, 12)))

# Plot both side by side
par(mfrow = c(1, 2))

# Additive seasonal factors (one year)
plot(seasonal_add_cycle, main = "Seasonal Factors (Additive) - One Year Cycle",
     ylab = "Seasonal Component", xlab = "Month", type = "l", col = "blue", lwd = 2,
     xaxt = "n")
axis(1, at = 1:12, labels = month.abb)
abline(h = 0, col = "red", lty = 2)

# Multiplicative seasonal factors (one year)
plot(seasonal_mult_cycle, main = "Seasonal Factors (Multiplicative) - One Year Cycle",
     ylab = "Seasonal Component", xlab = "Month", type = "l", col = "green", lwd = 2,
     xaxt = "n")
axis(1, at = 1:12, labels = month.abb)
abline(h = 1, col = "red", lty = 2)
```



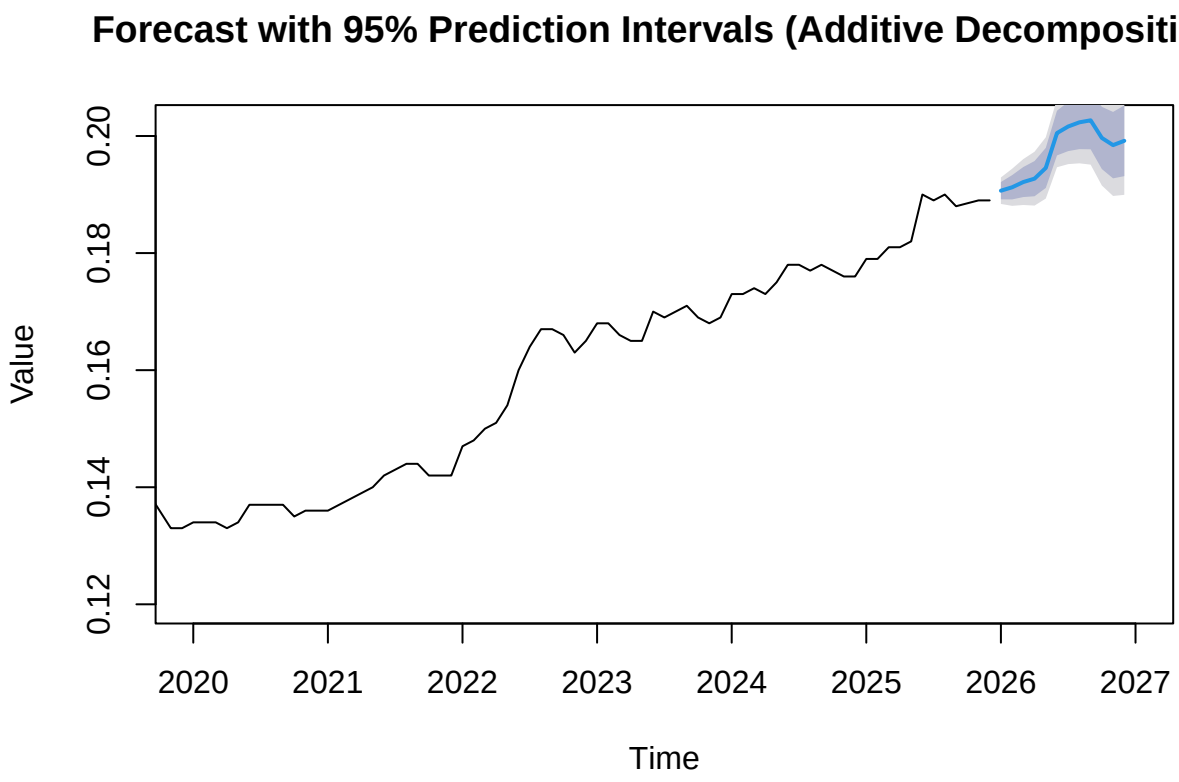
Both show a very similar yearly seasonal pattern which peaks in the summer months (June through September) and is lowest in the winter months. This would be consistent with real life as summer months mean people using more energy, and thus higher energy prices.

2f)

```r
# Forecast using additive decomposition

# Forecast the decomposed series
forecast_add <- forecast(decomp, h = 12)

# Plot the forecast with prediction intervals
par(mar = c(5, 4, 4, 2))
plot(forecast_add, main = "Forecast with 95% Prediction Intervals (Additive Decomposition)",
     ylab = "Value", xlab = "Time", xlim = c(2020, end(forecast_add$x)[1] + 2),
     ylim = c(0.12, 0.202))
```



After analyzing different models and types of decompositions, a non-linear trend with a quadratic component mixed with an additive decomposition is the most relevant model for this data.

Part 3

After choosing the preferred model and forecast, there is a clear upward trend that is slightly quadratic as well as a relevant seasonality trend which includes increased energy prices in the summer and lower energy prices in the winter. However, there are still some dynamics in the random component after the decomposition, likely due to cycles. After running a forecast 12 months into the future, energy prices are predicted to continue to increase while still following the seasonal pattern. For the future, it would be better to be able to further dive into the seasonal trends and the dynamics that are present in the ACF and PACF after the decomposition.

Part 4

I got my data from FRED (linked below)

https://fred.stlouisfed.org/series/APU000072610